새 **Cisco Nexus 92300YC** 스위치를 구성합니다

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새 Cisco Nexus 92300YC 스위치를 구성합니다

새 Cisco Nexus 92300YC 스위치를 구성합니다

이 장에 설명된 단계를 완료하여 새 Nexus 92300YC 스위치를 구성할 수 있습니다.

ONTAP 9.6 이상을 실행하는 시스템에 Nexus 92300YC 스위치를 설치하려면 스위치가 관리 인터페이스를 통해 통신할 수 있도록 IP 주소 및 구성을 설정해야 합니다. 그런 다음 NX-OS 소프트웨어 및 참조 구성 파일(RCF)을 설치할 수 있습니다. 이 절차는 컨트롤러를 추가하기 전에 Nexus 92300YC 스위치를 준비하는 데 사용됩니다.

이 절차의 예에서는 다음 스위치 및 노드 명명법을 사용합니다.

- Nexus 92300YC 스위치 이름은 CS1, CS2입니다.
- 이 절차에 사용된 예는 두 번째 스위치 _ * CS2 *._에서 업그레이드를 시작합니다
- 클러스터 LIF 이름은 node1의 node1_clus1 및 node1_clus2, node2의 node2_clus1 및 node2_clus2 입니다.
- IPspace 이름은 Cluster입니다.
- 'cluster1:: *>' 프롬프트는 클러스터의 이름을 나타냅니다.
- 각 노드의 클러스터 포트 이름은 e0a, e0b 입니다.

  플랫폼에서 지원되는 실제 클러스터 포트는 _Hardware Universe_를 참조하십시오.

  "NetApp Hardware Universe를 참조하십시오"

- Nexus 92300YC 스위치에 지원되는 ISL(Inter-Switch Link)은 포트 1/65 및 1/66입니다.
- Nexus 92300YC 스위치에서 지원되는 노드 연결은 포트 1/1 ~ 1/66입니다.
- 이 절차의 예에서는 2개의 노드를 사용하지만 하나의 클러스터에 최대 24개의 노드를 포함할 수 있습니다.

Cisco Nexus 92300YC 스위치의 초기 설치

이 절차를 사용하여 Cisco Nexus 92300YC 스위치의 초기 설치를 수행할 수 있습니다.

스위치에 해당하는 NetApp Cisco NX-OS 소프트웨어를 다운로드하려면 NetApp Support 사이트 에서 다운로드할 수 있습니다 "mysupport.netapp.com"

NX-OS는 Cisco Systems에서 제공하는 Nexus 시리즈 이더넷 스위치 및 MDS 시리즈 FC(Fibre Channel) 스토리지 영역 네트워크 스위치용 네트워크 운영 체제입니다.

이 절차에서는 스위치를 설치하고 실행하기 위한 프로세스를 간략하게 설명합니다.

단계

1. 직렬 포트를 선택한 호스트 또는 직렬 포트에 연결합니다.
2. 관리 포트(스위치의 포트 측 아님)를 SFTP 서버가 있는 동일한 네트워크에 연결합니다.
3. 콘솔에서 호스트 측 직렬 설정을 설정합니다.
   - 9600보드

- 8 데이터 비트
- 1 정지 비트
- 패리티: 없음
- 흐름 제어: 없음

4. 처음으로 부팅하거나 실행 중인 구성을 지운 후 재부팅하면 부팅 주기에서 Nexus 92300YC 스위치 루프가 됩니다. Auto Provisioning의 전원을 중단하려면 * yes * 를 입력하여 이 주기를 중단합니다. 그러면 시스템 관리자 계정 설정이 표시됩니다.

```
$ VDC-1 %$ %POAP-2-POAP_INFO:   - Abort Power On Auto Provisioning [yes
- continue with normal setup, skip - bypass password and basic
configuration, no - continue with Power On Auto Provisioning]
(yes/skip/no)[no]:
**yes**
Disabling POAP.......Disabling POAP
2019 Apr 10 00:36:17 switch %$ VDC-1 %$ poap: Rolling back, please
wait... (This may take 5-15 minutes)
```

```
---- System Admin Account Setup ----
```

```
 Do you want to enforce secure password standard (yes/no) [y]:
. 보안 암호 표준을 적용하려면 * y * 를 입력합니다.
```

```
Do you want to enforce secure password standard (yes/no) [y]: **y**
```

5. admin 사용자의 암호를 입력하고 확인합니다.

```
Enter the password for "admin":
Confirm the password for "admin":
```

6. 기본 시스템 구성 대화 상자를 엽니다.

```
This setup utility will guide you through the basic configuration of
the system. Setup configures only enough connectivity for management
of the system.

Please register Cisco Nexus9000 Family devices promptly with your
supplier. Failure to register may affect response times for initial
service calls. Nexus9000 devices must be registered to receive
entitled support services.

Press Enter at anytime to skip a dialog. Use ctrl-c at anytime
to skip the remaining dialogs.

Would you like to enter the basic configuration dialog (yes/no):
```

7. 다른 로그인 계정 만들기:

```
Create another login account (yes/no) [n]:
```

8. 읽기 전용 및 읽기-쓰기 SNMP 커뮤니티 문자열 구성:

```
Configure read-only SNMP community string (yes/no) [n]:

Configure read-write SNMP community string (yes/no) [n]:
```

9. 클러스터 스위치 이름 구성:

```
Enter the switch name : **cs2**
```

10. 대역외 관리 인터페이스 구성:

```
Continue with Out-of-band (mgmt0) management configuration? (yes/no)
[y]: **y**

Mgmt0 IPv4 address : 172.22.133.216

Mgmt0 IPv4 netmask : 255.255.224.0

Configure the default gateway? (yes/no) [y]: **y**

IPv4 address of the default gateway : 172.22.128.1
```

11. 고급 IP 옵션 구성:

```
Configure advanced IP options? (yes/no) [n]: **n**
```

12. 텔넷 서비스 구성:

```
Enable the telnet service? (yes/no) [n]: **n**
```

13. SSH 서비스 및 SSH 키 구성:

```
Enable the ssh service? (yes/no) [y]: **y**

       Type of ssh key you would like to generate (dsa/rsa) [rsa]:
**rsa**

       Number of rsa key bits <1024-2048> [1024]: **2048**
```

14. 기타 설정 구성:

```
Configure the ntp server? (yes/no) [n]: **n**

  Configure default interface layer (L3/L2) [L2]: **L2**

  Configure default switchport interface state (shut/noshut) [noshut]:
**noshut**

  Configure CoPP system profile (strict/moderate/lenient/dense)
[strict]: **strict**
```

15. 스위치 정보를 확인하고 구성을 저장합니다.

```
Would you like to edit the configuration? (yes/no) [n]: **n**

Use this configuration and save it? (yes/no) [y]: **y**

[########################################] 100%
Copy complete, now saving to disk (please wait)...
Copy complete.
```

NX-OS 소프트웨어를 설치합니다

이 절차를 사용하여 Nexus 92300YC 스위치에 NX-OS 소프트웨어를 설치할 수 있습니다.

단계

1. 클러스터 스위치를 관리 네트워크에 연결합니다.
2. "ping" 명령을 사용하여 NX-OS 소프트웨어 및 RCF를 호스팅하는 서버에 대한 연결을 확인합니다.

   이 예에서는 스위치가 IP 주소 172.19.2.1로 서버에 연결할 수 있는지 확인합니다.

   ```
   cs2# **ping 172.19.2.1**
   Pinging 172.19.2.1 with 0 bytes of data:

   Reply From 172.19.2.1: icmp_seq = 0. time= 5910 usec.
   ```

3. NX-OS 소프트웨어 및 EPLD 이미지를 Nexus 92300YC 스위치에 복사합니다.

```
cs2# **copy sftp: bootflash: vrf management**
Enter source filename: **/code/nxos.9.2.2.bin**
Enter hostname for the sftp server: **172.19.2.1**
Enter username: **user1**

Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/nxos.9.2.2.bin  /bootflash/nxos.9.2.2.bin
/code/nxos.9.2.2.bin  100% 1261MB   9.3MB/s   02:15
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.


cs2# **copy sftp: bootflash: vrf management**
Enter source filename: **/code/n9000-epld.9.2.2.img**
Enter hostname for the sftp server: **172.19.2.1**
Enter username: **user1**

Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/n9000-epld.9.2.2.img  /bootflash/n9000-epld.9.2.2.img
/code/n9000-epld.9.2.2.img  100%  161MB   9.5MB/s   00:16
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.
```

4. 실행 중인 NX-OS 소프트웨어 버전을 확인합니다.

```
cs2# **show version**
Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
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All rights reserved.
The copyrights to certain works contained in this software are
owned by other third parties and used and distributed under their own
licenses, such as open source.  This software is provided "as is," and
unless
otherwise stated, there is no warranty, express or implied, including
```

```
but not
limited to warranties of merchantability and fitness for a particular
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the GNU General Public License (GPL) version 2.0 or
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Lesser General Public License (LGPL) Version 2.1 or
Lesser General Public License (LGPL) Version 2.0.
A copy of each such license is available at
http://www.opensource.org/licenses/gpl-2.0.php and
http://opensource.org/licenses/gpl-3.0.html and
http://www.opensource.org/licenses/lgpl-2.1.php and
http://www.gnu.org/licenses/old-licenses/library.txt.

Software
  BIOS: version 05.31
  NXOS: version 9.2(1)
  BIOS compile time:  05/17/2018
  NXOS image file is: bootflash:///nxos.9.2.1.bin
  NXOS compile time:  7/17/2018 16:00:00 [07/18/2018 00:21:19]


Hardware
  cisco Nexus9000 C92300YC Chassis
  Intel(R) Xeon(R) CPU D-1526 @ 1.80GHz with 16337884 kB of memory.
  Processor Board ID FDO220329V5

  Device name: cs2
  bootflash:  115805356 kB
Kernel uptime is 0 day(s), 4 hour(s), 23 minute(s), 11 second(s)

Last reset at 271444 usecs after Wed Apr 10 00:25:32 2019
  Reason: Reset Requested by CLI command reload
  System version: 9.2(1)
  Service:

plugin
  Core Plugin, Ethernet Plugin

Active Package(s):

cs2#
```

5. NX-OS 이미지를 설치합니다.

   이미지 파일을 설치하면 스위치를 재부팅할 때마다 이미지 파일이 로드됩니다.

```
cs2# **install all nxos bootflash:nxos.9.2.2.bin**

Installer will perform compatibility check first. Please wait.
Installer is forced disruptive

Verifying image bootflash:/nxos.9.2.2.bin for boot variable "nxos".
[####################] 100% -- SUCCESS

Verifying image type.
[####################] 100% -- SUCCESS

Preparing "nxos" version info using image bootflash:/nxos.9.2.2.bin.
[####################] 100% -- SUCCESS

Preparing "bios" version info using image bootflash:/nxos.9.2.2.bin.
[####################] 100% -- SUCCESS

Performing module support checks.
[####################] 100% -- SUCCESS

Notifying services about system upgrade.
[####################] 100% -- SUCCESS



Compatibility check is done:
Module  bootable       Impact     Install-type  Reason
------  --------  --------------- ------------  ------
  1       yes      disruptive         reset     default upgrade is not
hitless



Images will be upgraded according to following table:

Module   Image         Running-Version(pri:alt            New-Version
Upg-Required
------  --------  --------------------------------------
------------------  ------------
  1       nxos                                  9.2(1)
9.2(2)         yes
  1       bios    v05.31(05/17/2018):v05.28(01/18/2018)
v05.33(09/08/2018)       yes


Switch will be reloaded for disruptive upgrade.
```

```
Do you want to continue with the installation (y/n)?  [n] **y**


Install is in progress, please wait.

Performing runtime checks.
[####################] 100% -- SUCCESS

Setting boot variables.
[####################] 100% -- SUCCESS

Performing configuration copy.
[####################] 100% -- SUCCESS

Module 1: Refreshing compact flash and upgrading bios/loader/bootrom.
Warning: please do not remove or power off the module at this time.
[####################] 100% -- SUCCESS
2019 Apr 10 04:59:35 cs2 %$ VDC-1 %$ %VMAN-2-ACTIVATION_STATE:
Successfully deactivated virtual service 'guestshell+'

Finishing the upgrade, switch will reboot in 10 seconds.
```

6. 스위치를 재부팅한 후 새로운 버전의 NX-OS 소프트웨어를 확인합니다.

   'How version'입니다

```
cs2# **show version**

Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
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The copyrights to certain works contained in this software are
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otherwise stated, there is no warranty, express or implied, including
but not
limited to warranties of merchantability and fitness for a particular
purpose.
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GNU General Public License (GPL) version 3.0  or the GNU
Lesser General Public License (LGPL) Version 2.1 or
Lesser General Public License (LGPL) Version 2.0.
A copy of each such license is available at
```

```
http://www.opensource.org/licenses/gpl-2.0.php and
http://opensource.org/licenses/gpl-3.0.html and
http://www.opensource.org/licenses/lgpl-2.1.php and
http://www.gnu.org/licenses/old-licenses/library.txt.

Software
  BIOS: version 05.33
  NXOS: version 9.2(2)
  BIOS compile time:  09/08/2018
  NXOS image file is: bootflash:///nxos.9.2.2.bin
  NXOS compile time:  11/4/2018 21:00:00 [11/05/2018 06:11:06]


Hardware
  cisco Nexus9000 C92300YC Chassis
  Intel(R) Xeon(R) CPU D-1526 @ 1.80GHz with 16337884 kB of memory.
  Processor Board ID FDO220329V5

  Device name: cs2
  bootflash:  115805356 kB
  Kernel uptime is 0 day(s), 0 hour(s), 3 minute(s), 52 second(s)

Last reset at 182004 usecs after Wed Apr 10 04:59:48 2019
  Reason: Reset due to upgrade
  System version: 9.2(1)
  Service:

plugin
  Core Plugin, Ethernet Plugin

Active Package(s):
```

7. EPLD 이미지를 업그레이드하고 스위치를 reboot한다.

```
cs2# **show version module 1 epld**

EPLD Device                      Version
---------------------------------------
MI FPGA                           0x7
IO FPGA                           0x17
MI FPGA2                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2

cs2# **install epld bootflash:n9000-epld.9.2.2.img module 1**
Compatibility check:
Module        Type         Upgradable        Impact   Reason
------  -----------------  ----------    ----------   ------
     1            SUP           Yes       disruptive  Module Upgradable

Retrieving EPLD versions.... Please wait.
Images will be upgraded according to following table:
Module  Type   EPLD              Running-Version   New-Version  Upg-
Required
------  ----  -------------      ---------------   -----------
------------
     1   SUP  MI FPGA                   0x07        0x07             No
     1   SUP  IO FPGA                   0x17        0x19            Yes
     1   SUP  MI FPGA2                  0x02        0x02             No
The above modules require upgrade.
The switch will be reloaded at the end of the upgrade
Do you want to continue (y/n) ?  [n] **y**

Proceeding to upgrade Modules.

Starting Module 1 EPLD Upgrade

Module 1 : IO FPGA [Programming] : 100.00% (     64 of      64 sectors)
Module 1 EPLD upgrade is successful.
Module        Type  Upgrade-Result
------  ------------------  --------------
     1         SUP         Success


EPLDs upgraded.

Module 1 EPLD upgrade is successful.
```

8. 스위치 재부팅 후 다시 로그인하여 새 버전의 EPLD가 성공적으로 로드되었는지 확인합니다.

```
cs2# **show version module 1 epld**

EPLD Device                      Version
---------------------------------------
MI FPGA                           0x7
IO FPGA                           0x19
MI FPGA2                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
```

RCF(Reference Configuration File) 설치

Nexus 92300YC 스위치를 처음 설정한 후 RCF를 설치할 수 있습니다.

단계

1. 클러스터 스위치를 관리 네트워크에 연결합니다.
2. RCF를 호스팅하는 서버에 대한 연결을 확인하려면 ping 명령을 사용합니다.

   이 예에서는 스위치가 IP 주소 172.19.2.1로 서버에 연결할 수 있는지 확인합니다.

```
cs2# **ping 172.19.2.1**
Pinging 172.19.2.1 with 0 bytes of data:

Reply From 172.19.2.1: icmp_seq = 0. time= 5910 usec.
```

3. RCF를 Nexus 92300YC 스위치에 복사합니다.

```
cs2# **copy sftp: bootflash: vrf management**
Enter source filename: **/code/Nexus_92300YC_RCF_v1.0.2.txt**
Enter hostname for the sftp server: **172.19.2.1**
Enter username: **user1**

Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/Nexus_92300YC_RCF_v1.0.2.txt
/bootflash/nxos.9.2.2.bin
/code/Nexus_92300YC_R  100% 9687   530.2KB/s   00:00
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.
```

4. RCF를 스위치의 running-config와 병합합니다.

```
cs2# **copy bootflash:Nexus_92300YC_RCF_v1.0.2.txt running-config**

Disabling ssh: as its enabled right now:
 generating ecdsa key(521 bits)......
generated ecdsa key

Enabling ssh: as it has been disabled
 this command enables edge port type (portfast) by default on all
interfaces. You
 should now disable edge port type (portfast) explicitly on switched
ports leading to hubs,
 switches and bridges as they may create temporary bridging loops.

Edge port type (portfast) should only be enabled on ports connected to a
single
 host. Connecting hubs, concentrators, switches, bridges, etc...  to
this
 interface when edge port type (portfast) is enabled, can cause
temporary bridging loops.
 Use with CAUTION

Edge Port Type (Portfast) has been configured on Ethernet1/1 but will
only
 have effect when the interface is in a non-trunking mode.

...

Copy complete, now saving to disk (please wait)...
Copy complete.
```

5. 스위치에서 RCF가 성공적으로 병합되었는지 확인합니다.

   'show running-config'를 선택합니다

```
cs2# **show running-config**
!Command: show running-config
!Running configuration last done at: Wed Apr 10 06:32:27 2019
!Time: Wed Apr 10 06:36:00 2019

version 9.2(2) Bios:version 05.33
switchname cs2
vdc cs2 id 1
  limit-resource vlan minimum 16 maximum 4094
  limit-resource vrf minimum 2 maximum 4096
  limit-resource port-channel minimum 0 maximum 511
  limit-resource u4route-mem minimum 248 maximum 248
  limit-resource u6route-mem minimum 96 maximum 96
  limit-resource m4route-mem minimum 58 maximum 58
  limit-resource m6route-mem minimum 8 maximum 8

feature lacp

no password strength-check
username admin password 5
$5$HY9Kk3F9$YdCZ8iQJ1RtoiEFa0sKP5IO/LNG1k9C4lSJfi5kesl
6  role network-admin
ssh key ecdsa 521

banner motd #
***********************************************************************
********
*
*
*  Nexus 92300YC Reference Configuration File (RCF) v1.0.2 (10-19-2018)
*
*
*
*  Ports 1/1  - 1/48: 10GbE Intra-Cluster Node Ports
*
*  Ports 1/49 - 1/64: 40/100GbE Intra-Cluster Node Ports
*
*  Ports 1/65 - 1/66: 40/100GbE Intra-Cluster ISL Ports
*
*
*
***********************************************************************
********
```

6. 스위치를 재부팅할 때 시작 구성이 되도록 실행 중인 구성을 저장합니다.

```
cs2# **copy running-config startup-config**

[#########################################] 100%
Copy complete, now saving to disk (please wait)...
Copy complete.
```

7. ONTAP 9.6P8 이상에서는 스위치 관련 로그 파일(시스템 클러스터 스위치 로그 설정 - 암호, 시스템 클러스터 - 스위치 로그 활성화 - 수집)을 수집하기 위한 CSHM ASUP 로그 수집 기능을 활성화합니다

```
cs2# **system cluster-switch log setup-password**
** Output example required here
cs2# **system cluster-switch log enable-collection**
** Output example required here too
```

8. 스위치를 재부팅하고 실행 중인 구성이 올바른지 확인합니다.

   다시 로드

```
cs2# **reload**

This command will reboot the system. (y/n)?  [n] **y**
```

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